

Zhiwei Li

AI ENGINEER · FULL STACK ENGINEER

Guangzhou, China

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Summary

- Software engineer with 11+ years of experience spanning full-stack development, mobile engineering, and machine learning; currently an AI Full Stack Engineer outsourced to HSBC Bank.
- Strong AI/ML fundamentals: trained GPT-2 language models (124M–760M parameters) from scratch on H200, H100, and AMD MI300X GPUs; authored SEC-EDGAR-GPT, a paper on training a language model on SEC EDGAR filings.
- Deep, hands-on experience with Java, Spring, Python, and React, plus cloud platforms including AWS, Azure, and Alibaba Cloud; ranked top 6% in internal Copilot usage at HSBC.
- Self-driven learner: IELTS Academic 6.5, 400+ original technical blog posts, and 10+ open-source projects with 500+ commits each on GitHub.

Education

Guangdong University of Foreign Studies — National Higher Education Self-Study Examination

Guangzhou, China

ASSOCIATE DEGREE IN COMPUTER APPLICATION (IN PROGRESS)

2022 – Present

- Passed 13 written courses, including Advanced Mathematics, Data Structures, Operating Systems, Java Programming, Databases, and Computer Organization.
- Passed 3 practical exams in programming, computer fundamentals, and database applications.

Beijing Forestry University

Beijing, China

BACHELOR'S IN DIGITAL MEDIA ARTS (INCOMPLETE)

2013 – 2014

- Completed 13 courses in programming and digital media; built a campus-assistant Android app adopted by thousands of students; won third prize in a hackathon.
- Awarded First Prize in the 2014 Blue Bridge Cup programming contest.

Work Experience

TEKsystems

Guangzhou, China

AI FULL STACK ENGINEER (AT HSBC BANK)

Feb 2025 – Present

- Develop finance transformation features (import, validation, export, and approval workflows) for a module of HSBC's finance ledger platform using Java, Spring, IBM Db2, and Angular.
- Drive the full development lifecycle from local development to UAT and production; automated releases with Ansible and Jenkins and assisted in upgrading Angular and decommissioning legacy WebSphere applications.
- Use the Copilot API to auto-generate 70 test cases covering Spring filters, Python unittest, and prompt engineering; ranked top 6% in internal Copilot usage and earned a Contribution Award for the AIPlayer project.

Self-Employed

Guangzhou, China

INDEPENDENT AI / SOFTWARE ENGINEER

Jul 2023 – Jan 2025

- Trained GPT-2 124M from scratch on 1.55B tokens of SEC EDGAR financial filings on a single RTX 4070, and scaled to 124M–760M parameters on H200, H100, and AMD MI300X GPUs through a nanoGPT fork (45 commits).
- Built an AI-powered story bot with Python, Flask, React, and the Claude API on AWS, monitored with Prometheus and the ELK stack.
- Reimplemented core components of 30 machine learning projects from PyTorch, TensorFlow, and Coursera coursework; completed the Machine Learning and Deep Learning Specializations.

Farben Information

Guangzhou, China

BACKEND ENGINEER (AT HSBC PAYME)

Nov 2022 – Jul 2023

- Developed the Auto Top Up feature for PayMe, HSBC's mobile payment service in Hong Kong, using Java, Spring, and Kafka; monitored post-payment events from Azure EventHub.
- Led a cloud migration to AWS following official AWS training; refactored APIs for request header-based routing and deployed microservices to the new infrastructure.

Beyondsoft

BACKEND ENGINEER (AT DBS BANK)

Guangzhou, China

Dec 2021 – Nov 2022

- Built equity trading microservices for DBS Client Connect, including pre-trade checks, order placement, and Avaloq API integration; refined stock code search with an edit-distance algorithm.
- Developed mutual fund and investment portfolio microservices for DBS DigiBank CN; automated test generation for Karate and analyzed performance on Pivotal Cloud Foundry.

Square Root Technology

FOUNDER & FULL STACK ENGINEER

Beijing, China

Jul 2016 – Dec 2019

- Founded and operated Fun Live, a knowledge live-streaming platform that reached 30,000 users; personally led engineering across 3 web clients and 1 server (2,000 commits) using PHP, Vue, CodeIgniter, MySQL, and Redis.
- Delivered 50+ software consulting projects (websites, apps, WeChat Mini Programs) with approximately 3M RMB in revenue; responsible for negotiation, team management, and development.

LeanCloud

SOFTWARE ENGINEER

Beijing, China

Jul 2014 – Nov 2015

- Developed LeanCloud's Objective-C and Java SDKs and the LeanChat iOS/Android demo apps for the instant messaging platform.

Selected Projects

AI Research — Language Model Training

Open Source

SEC-EDGAR-GPT

2025 – 2026

- Trained a GPT-2 124M language model from scratch on 1.55B tokens of SEC EDGAR financial filings (10-K, 10-Q, 8-K); converged to validation loss 2.28 in 8 hours on a single RTX 4070.
- Authored a technical paper describing the approach; released code on GitHub and the trained model on Hugging Face.

AI Research — Model Training

Open Source

NANO GPT FORK: MULTI-GPU MODEL TRAINING

2025 – 2026

- Trained GPT-2 models (124M–760M parameters) on FineWeb and OpenWebText datasets across H200, H100, and AMD MI300X GPUs; built 6 dataset pipelines and training configs for different GPU budgets.
- Tech: Python, PyTorch, CUDA, ROCm, FlashAttention, HuggingFace.

AI-Powered Technical Blog Platform

Open Source

JEKYLL-AI-BLOG

2023 – Present

- Built an AI-powered blogging platform with LLM-based translation into 9+ languages, Google Cloud text-to-speech, XeLaTeX PDF/EPUB export, and GitHub Actions CI/CD.
- Hosts 400+ original posts and 8,000 AI study notes with 70,000 page views per month; Tech: Jekyll, Python, DeepSeek, Mistral.

Full Stack Development — Entrepreneurial Startup

Beijing, China

FUN LIVE

2016 – 2017

- Founded and operated a knowledge live-streaming platform that reached 30,000 users and hosted 80 lectures; users paid to attend live sessions, rewarded lecturers, or watched playbacks, with seamless WeChat notifications.
- Full-stack ownership across 3 web clients and 1 server (2,000 commits, later open-sourced on GitHub); Tech: PHP, Vue, CodeIgniter, MySQL, Redis, LeanCloud, Alibaba Cloud, RTMP Streaming.

Skills

- **Programming:** Java, Python, PHP, TypeScript, JavaScript, Objective-C, Swift, C/C++, SQL, HTML/CSS
- **Frameworks:** Spring, Spring Boot, Flask, React, Vue, Angular, Laravel, CodeIgniter, Android SDK, iOS SDK
- **AI / ML:** PyTorch, CUDA, ROCm, Hugging Face, llama.cpp, nanoGPT, RAG, Prompt Engineering, LLM APIs (OpenAI, Claude, Copilot)
- **Data & Infrastructure:** MySQL, PostgreSQL, IBM Db2, Redis, Kafka, Docker, Kubernetes, Nginx, Ansible, Jenkins, GitHub Actions
- **Cloud:** AWS, Azure, Alibaba Cloud, Pivotal Cloud Foundry, Cloudflare
- **Languages:** Chinese (native), English (IELTS Academic 6.5), Cantonese

Certificates & Awards

- 2025 **IELTS Academic, Score 6.5**, British Council
- 2023 **Machine Learning Specialization**, DeepLearning.AI & Stanford University
- 2023 **Deep Learning Specialization**, DeepLearning.AI
- 2014 **First Prize, Blue Bridge Cup Programming Contest**
- 2011 **Top 300, Guangdong NOIP**, National Olympiad in Informatics in Provinces